

Reducing Refrigerator Power During Peak Hours

by Greg Ward

Refrigerators consume about 14% of the total energy in the average U.S. household. While this may not be a very large figure, it is large enough that if most homes could cycle off their refrigerators during peak hours, we could significantly forestall the construction of new power plants. I propose a simple, low-cost modification to a standard refrigerator/freezer unit that achieves this goal. The basic requirement is the addition of a small amount (2 gallons or so) of phase change medium near the freezer coils, and some additional control logic.

Typical frost-free refrigerator/freezer units have a single evaporator coil in the freezer section, and a small vent that allows some air to circulate between the freezer and refrigerator sections. A single thermostat controls the cycling of the system to maintain each section in the desirable temperature range. For the refrigerator section, this range is 33°F–40°F. For the freezer section, it is somewhat wider, ideally -13°F–14°F, although anything below freezing is acceptable for a short period. (The freezer temperature affects mainly the softness of any ice cream in the freezer.) Simply cutting off power to the unit during peak demand hours could raise the temperature dangerously in both sections. Therefore, I propose the following modifications:

- Install a quantity of liquid with a lower freeze/melt temperature than water in the freezer section
- Install a timer or similar control on the compressor and condenser fan (if any) to disable active cooling during peak hours.

Do This at Home

In my test, I placed a 1:4 mixture of Sierra nontoxic antifreeze and water in



The author's freezer compartment with sub-zero ice bags on the top shelf.

two 1-gallon bags and set them on the middle shelf of the freezer (see photo above). To disengage the compressor while allowing the internal fan to continue circulating air between the two sections, I installed an electronic timer. This was set so that the compressor was off and the unit was in passive mode during the designated peak hours of 1PM to 7PM each day. In a more sophisticated realization, the passive hours could be varied slightly at random, or based on the frequency with which the door was opened, or the line voltage, or some combination of factors. Likewise, the low-temperature ice would be placed out of the way, adjacent to the evaporator coils or between sections, in such a way that the internal fan would not need to operate continuously in passive mode. Any commercial product would probably include an automatic override to ensure that the temperature in both sections never rose dangerously high.

This simple prototype serves as a proof of concept for the idea, and illustrates the parameters within which these modifications can succeed. To understand the effect of the modifications I made to a standard refrigerator, I compared the energy use of the standard refrigerator with that of the modified unit (see Figures 1 and 2).

As Figure 2 shows, the freezer section gradually climbs from its default temperature to about 28°F during the six hours spent in passive mode, while the refrigerator section maintains a more constant temperature, rising to a maximum of 41°F at the end of the peak period. I could alter these ranges by reducing the length of the passive period, decreasing the default temperature of the freezer, or using more subzero ice. The type of ice employed is also important. Mixing antifreeze into water doesn't achieve the same effect as one might get from using a different phase change medium—one that would not rise above a certain temperature until

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it had completely liquefied. Due to fractional freezing, an antifreeze mix will melt continuously above a certain temperature. This provides a coarser control over temperature than one would get with a pure phase change medium.

I also compared the power consumption of the standard unit with that of the modified unit (see Figure 3). The standard unit's energy use is constant at around 51 watts. The modified unit drops down to 5 watts during peak hours, rises to an average of 100 watts for an equivalent number of hours, and then drops to an average of 50 watts once the freezer temperature has been restored (and the subzero ice has refrozen). The overall average power consumption of the modified unit is 56 watts, or 10% higher than the overall average power consumption of the standard unit.

Does It Work?

While the behavior of my prototype modification is less than ideal, I believe the results adequately demonstrate the potential of this approach. Even with the 25% passive running period, the freezer section was maintained below freezing and the refrigerator section was kept within safe tolerances. A shorter passive period, or one where the compressor was allowed to run only briefly, would have kept our ice cream harder. More sophisticated control logic would not add to the cost of a refrigerator that already employs microprocessor control, and it would solve most of the problems outlined above. The cost of the phase change medium is nominal, and it could be situated better than I have done in this limited proof of concept. **He**

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Temperature Profile for Standard Unit

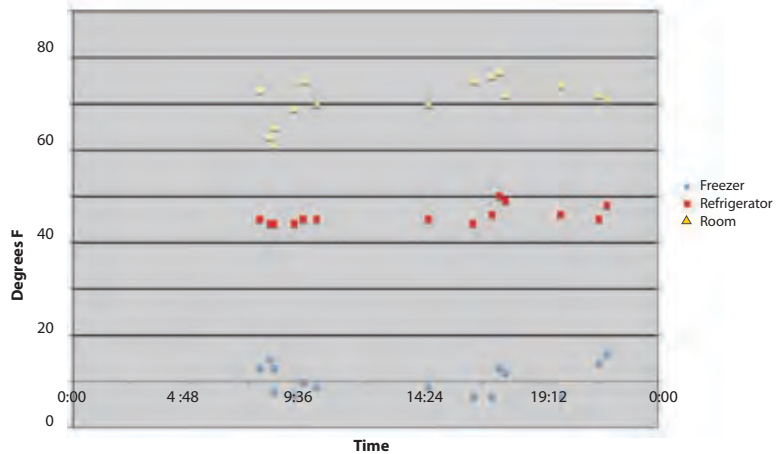


Figure 1. To understand the effect of the modifications I made to a standard refrigerator, I compared the energy use of the standard refrigerator with that of the modified unit

Temperature Profile for Modified Unit

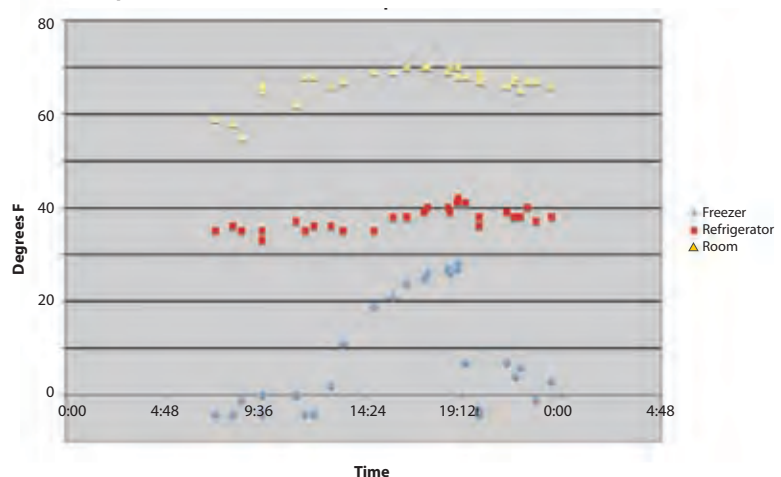


Figure 2. In the modified unit, the freezer section gradually climbs from its default temperature to about 28°F during the six hours spent in passive mode, while the refrigerator section maintains a more constant temperature, rising to a maximum of 41°F at the end of the peak period.

Power Consumption for Standard and Modified Units

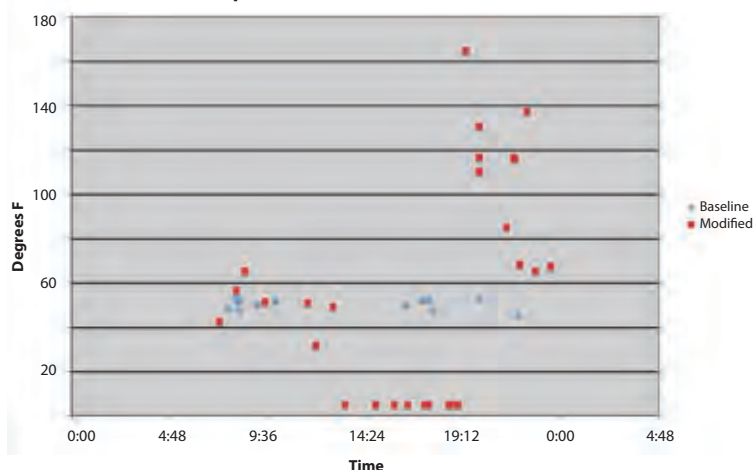


Figure 3. The overall average power consumption of the modified unit is 56 watts, or 10% higher than the overall average power consumption of the standard unit. While the behavior of my prototype modification is less than ideal, I believe the results adequately demonstrate the potential of this approach.